

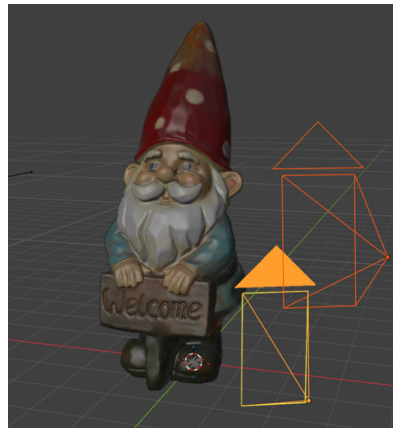
NTUST course: Computer Vision and Applications (CI5336701, 2026 Spring)

Homework#4: Reconstruct 3D from two images

Date Due : 2026. May. 26th, PM11:59 (2.0 weeks)

Description :

1. Write a program for reconstructing 3D points from sequential image pairs. The algorithm should extract laser lines, identify corresponding points between the pairs, and compute the 3D coordinates. Note that the intrinsic (K) and extrinsic (RT) matrices for each image pair are already provided. (choose your tools, ex. C++/C, python, openCV, or Matlab).
2. As shown in the figure, the cameras are in a vertical configuration. The epipolar constraint can be established by calculating the Fundamental matrix (F) directly from the individual K and RT matrices. This constraint is then used to find corresponding points between the image pairs to obtain the 3D coordinates.



3. Please save the calculated 3D coordinates in an .xyz format, where each line represents a single coordinate point. Furthermore, utilize third-party software to validate the accuracy of your results. And please reject outliers, if their errors are greater than a specific threshold.
4. Deliverable: There are three items you should provide:
 - 1) Source code (and execution file, if any) in python, C++/C, Matlab, with simple comments,
 - 2) An output 3D file in YourID.xyz (3D point cloud, no color),No need to write report.

Please zip all your files, then, upload to moodle2 (<http://moodle2.ntust.edu.tw>) by date due.

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